Theoretical Feasibility Evaluation of LoRa Communication for Wildfire Detection Networks

by

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This thesis looks at whether LoRa can realistically be used for a wildfire detection sensor network. The work builds on our senior capstone project, where we built and tested a small LoRa network and communicated over at least 500 m with about 97% packet delivery. I used those results as a starting point to look more closely at communication range, reliability, latency, and energy use. The calculations showed that the range achieved during our testing may change quite a bit in environments with vegetation and other obstructions. Adding relay links also reduces the overall chance of a packet, making it all the way to the receiver. For response time, I found that the 60-second transmission interval matters much more than the actual LoRa packet airtime. Power was another interesting part of the analysis because the LoRa transmission itself uses relatively little power compared with the MQ-2 smoke sensor and its heated sensing element. Based on these results, LoRa seems like a reasonable communication option for this type of network, although its performance would still depend heavily on the environment, node placement, and the sensing hardware being used.

Keywords: LoRa, wildfire detection, wireless sensor networks, forest propagation, link budget, low-power communication
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Theoretical Feasibility Evaluation of LoRa Communication for Wildfire Detection Networks

Gaurav Dange

***Abstract*—This thesis looks at whether LoRa can realistically be used for a wildfire-detection sensor network. The work builds on our senior capstone project, where we built and tested a small LoRa network and were able to communicate over at least 500 m with about 97% packet delivery. I used those results as a starting point to look more closely at communication range, reliability, latency, and energy use. The calculations showed that the range achieved during our testing may change quite a bit in environments with vegetation and other obstructions. Adding relay links also reduces the overall chance of a packet making it all the way to the receiver. For response time, I found that the 60-second transmission interval matters much more than the actual LoRa packet airtime. Power was another interesting part of the analysis because the LoRa transmission itself uses relatively little power compared with the MQ-2 smoke sensor and its heated sensing element. Based on these results, LoRa seems like a reasonable communication option for this type of network, although its performance would still depend heavily on the environment, node placement, and the sensing hardware being used.**

***Index Terms*—LoRa, wildfire detection, wireless sensor networks, forest propagation, link budget, low-power communication**

I. Introduction

WILDFIRES can spread quickly, especially in remote areas where they may not be noticed right away. Detecting a fire earlier gives emergency responders more time before it grows and spreads further. There are already several ways of monitoring wildfires, including satellites, cameras, unmanned aerial vehicles (UAVs), and ground-based sensor networks [1]. Each method works differently and has its own limitations. Satellites, for example, can monitor very large areas, but clouds, image resolution, and the time between observations can affect what they detect. Cameras are also useful, but they usually depend on visible smoke or fire being present [1].

For our project, we were interested in what could happen if sensors were placed directly in areas where wildfires are more likely to occur. These sensors could measure things like temperature, humidity, and smoke instead of relying only on something being visually detected. Similar systems have already been studied. Sendra et al., for example, used LoRa nodes to collect temperature, relative humidity, wind speed, and CO_2 measurements in rural areas [2]. The idea is that several sensor nodes could take measurements at different locations and send that information back to one receiver.

Communication becomes an important part of the system when these sensors are spread across a large or remote area. Running wires between nodes would not be practical in a forest, and frequently replacing batteries would also be difficult. The nodes therefore need a way to send small amounts of sensor data over useful distances without consuming too much power. Reliability and response time also matter because losing too many packets or receiving data too late would make the system less useful for detecting a possible fire.

We used LoRa in the capstone project mainly because we needed the sensor nodes to communicate over longer distances without using a large amount of power. Since each node only sends a small amount of sensor data, a high data rate was not really necessary for the system. LoRa also gave us some flexibility through settings such as spreading factor, bandwidth, coding rate, and transmit power [3], [4]. Depending on how these are set, the communication range, reliability, and the time needed to send a packet can change.

This thesis builds directly on the LoRa wildfire-detection system developed during our senior capstone project. The system used sensor transmitter nodes, relay nodes, and a receiver. Sensor data could either be sent directly to the receiver or forwarded through a relay when needed. During outdoor testing, we were able to communicate over at least 500 m and measured a packet-delivery ratio of approximately 97%. These results showed that the prototype could work, but it was only tested under a limited set of environmental and network conditions.

The thesis takes that system and looks at it from a more analytical point of view rather than redesigning the hardware. I use the results from the capstone project as a starting point and look at how distance, environmental signal loss, relay nodes, transmission timing, and energy consumption could affect its performance. The primary research question is:

> *Under what environmental and network conditions can LoRa provide reliable, timely, and energy-efficient communication for wildfire detection?*

To answer this question, the measurements from the capstone project are compared with theoretical calculations and results reported in previous LoRa research. The 500 m range and approximately 97% packet delivery are used as reference points rather than fixed limits for the system.

II. Reference System

This thesis uses the LoRa wildfire-detection system developed during the senior capstone project as the starting point for the analysis. The system included sensor nodes, relay nodes, and a receiver. Depending on the node's location, sensor data could either be sent directly to the receiver or forwarded

TABLE I: Reference-System Communication Parameters

Parameter	Baseline Value
Operating frequency	915 MHz
Spreading factor	SF7
Bandwidth	125 kHz
Transmit Power	17 dbm
Coding rate	4/5
Transmission interval	approximately 60 s
Packet forwarding limit	TTL = 2
Demonstrated distance	at least 500 m
Observed packet-delivery ratio	approximately 97%

through a relay. The main communication settings and the results from the capstone testing are shown in Table I.

These values are used as the baseline for the theoretical analysis in this thesis. During the capstone testing, the system successfully communicated over at least 500 m and achieved a packet-delivery ratio of approximately 97% under the tested conditions.

III. LoRa Communication and Related Work

LoRa is a low-power wireless technology that can send small amounts of data over relatively long distances. One advantage of LoRa is that its communication settings can be adjusted based on the system's needs. Parameters such as spreading factor, bandwidth, coding rate, and transmit power can affect the range, reliability, and transmission time of the communication link. For example, using a higher spreading factor can improve the receiver's ability to detect weaker signals, but it also increases the time needed to transmit a packet [3].

The environment is another major factor in how far a LoRa signal can travel. In a forest, the signal may have to pass through trees, vegetation, uneven terrain, and areas with different moisture levels. Ferreira et al. tested LoRa in forest, urban, and suburban environments and saw differences in received signal strength, signal-to-noise ratio, and packet delivery between the locations [4]. Because of this, the 500 m range we achieved during our capstone testing should be treated as a reference rather than something that would always be expected in a denser forest.

Previous work has also shown that LoRa can be used for environmental and wildfire-monitoring applications. Sendra et al. developed a LoRa-based network for monitoring temperature, humidity, wind speed, and CO_2 in rural areas [2]. Their work shows that LoRa can be used for this kind of distributed sensing, although the results can change depending on the environment and communication settings. In this thesis, we will use the capstone system as a reference and look at how some of these conditions could affect its performance.

IV. Theoretical Analysis

The performance of the reference system is evaluated using communication range, packet reliability, latency, and energy consumption. The capstone configuration in Table **??** is used as the starting point, and the analysis looks at how propagation loss, relay nodes, transmission timing, and energy use could affect the system.

A. Communication Range and Link Budget

To verify that two LoRa nodes can communicate successfully, we can calculate the expected power at the receiver. The power at the receiver can be compared with the receiver sensitivity to calculate the available link margin. A link budget used in previous LoRa research can be written as [5]

$$P_{RX} = P_{TX} + G_{TX} - L_{TX} - L_{path} + G_{RX} - L_{RX} \quad (1)$$

where P_{RX} and P_{TX} are the received and transmitted power, G_{TX} and G_{RX} are the antenna gains, L_{TX} and L_{RX} are the transmitter and receiver losses, and L_{path} is the propagation loss.

For a free-space reference case, the path loss can be calculated as [5], [6]

$$L_{FSPL} = 32.44 + 20\log_{10}(f) + 20\log_{10}(d) \quad (2)$$

where f is the operating frequency in megahertz and d is the distance between the transmitter and receiver in kilometers. This equation describes the path losses for an ideal free space environment.

A complete link-budget calculation would also need the antenna gains and the losses between the radio and antenna on both sides of the link. These values were not measured during the capstone project, so they are not included in the calculation here. They could be found from the antenna specifications or measured during future testing. Once those values are known, the link-budget equation can be used to estimate the received power and compare it with the receiver sensitivity.

B. Packet Reliability

Communication range alone does not show whether a wireless link is reliable. Packet-delivery ratio (PDR) is commonly used to evaluate LoRa communication performance and has been used in previous forest propagation studies together with RSSI and SNR [4]. For this analysis, PDR is defined as

$$PDR = \frac{N_{received}}{N_{transmitted}} \times 100\% \quad (3)$$

where $N_{received}$ is the number of packets successfully received and $N_{transmitted}$ is the total number of packets sent.

The capstone system achieved an observed PDR of approximately 97% during outdoor testing. To examine the effect of relay nodes, a simple multi-hop reliability model is used. If the packet must successfully cross every link in the route and the success of each link is assumed to be independent, the probability of successful end-to-end delivery can be written as

$$P_{route} = \prod_{i=1}^{H} P_i \quad (4)$$

where H is the number of communication links and P_i is the packet-success probability of link i. This is a simplified model and does not include retransmissions, duplicate packets, or failures that may be correlated between links.

C. Latency

Latency is important for a wildfire-detection network because sensor data needs to reach the receiver quickly enough to be useful. The airtime of a LoRa packet depends on things like spreading factor, bandwidth, coding rate, and packet size [7].

For this analysis, the nodes are assumed to send data about once every 60 s. If a change in temperature, humidity, or smoke happens right after a packet is sent, the new measurement could wait almost 60 s before the next transmission.

Relay nodes also add some delay because the packet has to be received and then transmitted again. A simple way to estimate this is

$$T_{route} \approx HT_{air} \tag{5}$$

where H is the number of communication links and T_{air} is the airtime of one packet. This ignores the small processing delay at each relay, but it gives an idea of how the communication time changes as more relay links are added.

For the reference calculation, a 96-byte packet using SF7, 125 kHz bandwidth, and coding rate 4/5 gives an airtime of approximately 0.164 s using the LoRa time-on-air calculation [7].

D. Energy Consumption

Energy consumption is important for this system because the sensor nodes may be placed in remote areas where replacing batteries often would not be practical. The amount of current used by a node also changes depending on what the node is doing. Previous LoRa energy models separate the operation of a sensor node into different states such as sleep, sensing, processing, transmission, and reception [8], [9].

Bouguera et al. describe the total energy used by a sensor node as

$$E_{Total} = E_{Sleep} + E_{Active} \tag{6}$$

where E_{Sleep} is the energy used while the node is sleeping and E_{Active} is the energy used while the node is active [8]. The active energy can be broken down further as

$$E_{Active} = E_{WU} + E_m + E_{proc} + E_{WUT} + E_{Tr} + E_R \tag{7}$$

where the terms represent wake-up, measurement, processing, transceiver wake-up, transmission, and reception energy [8].

Another way to look at the power usage is through the average current of the node. Casals et al. calculate this using

$$I_{avg} = \frac{1}{T_{Notif}} \sum_{i=1}^{N_{states}} T_i I_i \tag{8}$$

where T_i is the amount of time spent in each operating state and I_i is the current used during that state [9]. T_{Notif} represents the time between transmissions. For the reference system, packets are sent approximately once every 60 s.

Battery lifetime can then be estimated using

$$T_{lifetime} = \frac{C_{battery}}{I_{avg}} \tag{9}$$

where $C_{battery}$ is the battery capacity and I_{avg} is the average current [9]. This gives a theoretical estimate of battery life and does not include all of the losses that may occur in a real system.

For this thesis, the current consumption of the ESP32, RFM95W, BME280, and MQ-2 is taken from manufacturer datasheets. These values are then used to compare how much power each part of the sensor node requires and to estimate the energy requirement of the system.

V. Results and Discussion

The first calculation uses the 500 m communication distance achieved during the capstone project. Using 915 MHz and 0.5 km, the free-space path loss is

$$L_{FSPL} = 32.44 + 20\log_{10}(915) + 20\log_{10}(0.5) \tag{10}$$

which gives approximately 85.65 dB. This is an ideal free-space value and does not include additional loss from vegetation, terrain, or other obstructions [5], [6]. The actual path loss in a forest would therefore be higher.

Packet reliability also changes when relay nodes are added. The capstone system achieved approximately 97% packet delivery. If each link is assumed to have the same 97% success rate, a two-link route gives

$$P_{route} = (0.97)^2 = 0.9409 \tag{11}$$

or about 94.1%. For three links,

$$P_{route} = (0.97)^3 \approx 0.9127 \tag{12}$$

or about 91.3%. This shows that route reliability can decrease as more relay links are added, even when each individual link performs well.

The transmission interval has a much larger effect on the response time than the LoRa airtime. Using an airtime of approximately 0.164 s as an example, a direct transmission would take about 0.164 s of airtime. With one relay, the packet has to be transmitted twice, giving about 0.328 s. With two relays, the total airtime would be about 0.492 s. These values do not include the processing time at each relay.

Even with multiple relay links, the communication time is still much smaller than the possible 60 s wait for the next sensor transmission. Because of this, changing how often the nodes send data would have a much larger effect on the overall response time than the LoRa airtime itself.

Energy consumption was also considered for the sensor node. The MQ-2 requires a heated sensing element with a maximum heater power of approximately 950 mW [10]. The BME280 requires much less power, while the ESP32 current depends on its operating state [11], [12].

For the RFM95W, the manufacturer reports a typical transmit current of approximately 90 mA at 17 dBm [13]. Using a 3.3 V supply and the calculated airtime of approximately

0.164 s, the average LoRa transmit power over the 60-second analysis interval is

$$P_{LoRa,avg} = \frac{(3.3)(0.090)(0.164)}{60} \approx 0.81 \text{ mW}. \quad (13)$$

Compared with the MQ-2 heater, the average LoRa transmit power is very small. This suggests that the sensing hardware may have a larger effect on the total node power than the LoRa transmission itself.

Since the battery and solar system have not yet been selected, battery capacity is kept as a variable. If the usable battery energy is $E_{battery}$ and the average node power is P_{node}, the battery-only operating time can be written as

$$T_{battery} = \frac{E_{battery}}{P_{node}}. \quad (14)$$

A final operating-life estimate cannot be made until the battery capacity and solar-charging system are defined. For this system, reducing the power used by the sensing hardware may be more important than only reducing LoRa transmit power.

VI. Limitations

The analysis in this thesis is based on a limited set of measurements from the original capstone project and several theoretical models. The 500 m communication test and approximately 97% packet-delivery ratio were measured under specific outdoor conditions and may not represent performance in a dense forest, wet vegetation, uneven terrain, or other environments.

The path-loss value used here is based on free space, so it does not fully represent what the signal would experience in a forest. Trees, terrain, and other obstacles could add more signal loss. The multi-hop reliability calculation is also simplified because each communication link is assumed to work independently. In an actual network, retransmissions, interference, duplicate packets, or the same environmental condition affecting several links could change the reliability. A complete received-power calculation was also not done because the antenna gains and RF losses were not measured during the capstone project. Because of this, the range analysis uses free-space path loss as a reference instead of trying to predict an exact received-power value.

The energy analysis also depends on operating times and current-consumption values obtained from component datasheets and the original capstone system. Actual battery life may differ because of factors such as regulator losses, battery condition, temperature, and changes in node activity. These limitations mean that the calculated results should be treated as estimates of system behavior rather than exact predictions of field performance.

VII. Conclusion

This thesis looked at whether LoRa communication can be realistically used for a wildfire-detection sensor network. The system developed during the capstone project was used as the main reference, and the analysis focused on communication range, packet reliability, latency, and energy consumption.

The capstone system was able to communicate over at least 500 m and achieved approximately 97% packet delivery during testing. However, these results were achieved under specific outdoor conditions. The calculations show that things like vegetation, terrain, and the number of relay nodes can change how well the network performs. As more relay links are added, the overall chance of a packet reaching the receiver can also decrease.

The latency analysis showed that the 60-second transmission interval has a much bigger effect on the response time than the actual LoRa packet airtime. The energy analysis also showed that the LoRa transmission itself is not the biggest power requirement in the node. The MQ-2 smoke sensor uses much more power because of its heated sensing element, so the sensing hardware has to be considered just as much as the communication hardware.

From the work done in this thesis, LoRa still looks like a practical communication option for this type of network. The main limitation is that its actual performance will depend heavily on where the nodes are placed, what the environment looks like, and what sensing hardware is used. The next step would be to test the system in real forest conditions and compare those results with the theoretical calculations used here.

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